

**BANARAS HINDU UNIVERSITY**

B.A./B.Sc. (Hons.) Semester III Examination 2025-26 (NEP)

**MATHEMATICS****Paper Code: MATMV31 : Python for Mathematical Application**

Ref. No.: CECNF/N/107

Time: 2½ Hours

Maximum Marks: 70

*Instructions: Answer Five questions including Question No. 1 which is compulsory. All programs must take input from the user. Use proper syntax and indentation.*

**Question 1.**

(a) Let `setA={3,6,9}`, `setB={1,3,9}`. What will be the result of the snippet `print(setA | setB)`? [2]

(b) State whether the following Python statements are valid or invalid. If valid, write the output: [2]

I. `print("Hello", "Python")`

II. `print("Hello" + 123)`

III. `print("Python", '+', "Programming")`

IV. `print("ABC is a 'Technological' University")`

(c) What will be the values of the variables `x`, `y`, and `z` after execution of the following Python program? Assume that the user enters the value 5. [1]

```
temp = input("Enter a number: ")
x = temp * 5
y = int(temp) * 5
z = bool(temp * 0)
```

(d) Explain the difference between `del` and `clear()` in dictionary with an example. [1]

(e) When should we use a nested if statement? Illustrate your answer with the help of an example. [1]

(f) Which Python function can be used to add more than one element within an existing list? [1]

(g) What is meant by variable scope and function call? [2]

**Question 2.**

(a) Explain data types in Python. Also, write a program that accepts two numbers from the user, validates whether the inputs are numeric, and performs addition, subtraction, multiplication, and division. [5]

(b) Explain the difference between built-in functions and user-defined functions in Python. Write a program that finds whether a given character is present in a string or not. If it is present, it prints all the indices at which it appears. Do not use the built-in function to search for the character. [5]

**Question 3.**

- (a) What are conditional statements in Python? Explain the `if`, `if-else`, and `if-elif-else` statements with suitable examples. Write a program to enter the marks of a student in four subjects. Then calculate the total and aggregate, and display the grade obtained by the student (Distinction  $> 75\%$ , First Division  $\geq 60\%$  and  $< 75\%$ , Second Division  $\geq 50\%$  and  $< 60\%$ , Third Division  $\geq 40\%$  and  $< 50\%$ , else Fail). [5]
- (b) Write the differences between dictionaries and sets in Python. Write a Python program in which the user enters a dictionary and the program inverts the dictionary (makes keys values and vice versa). [5]

**Question 4.**

- (a) Differentiate between local and global variables in Python. Write a Python program that shows how a global variable can be accessed and modified inside a function. [4]
- (b) Write a Python program to accept a list of elements from the user and remove duplicate elements using a set. Display the updated list. [3]
- (c) Write a Python program using a user-defined function to swap two numbers entered by the user and display the values before and after swapping. [3]

**Question 5.**

- (a) What are keywords in Python? Write any three keywords and explain their uses with examples. Write a program that prompts the user to enter a string. The program calculates and displays the length of the string until the user enters "QUIT". [5]
- (b) What is a set in Python? Explain its characteristics. Write a Python program to accept two sets from the user and find their union, intersection, difference, and symmetric difference. [5]

**Question 6.**

- (a) Explain functions, parameters, and return values in Python. Write a Python program using a user-defined function to calculate the factorial of a number entered by the user. Validate that the input is a non-negative integer. [5]
- (b) Explain tuples in Python and mention at least two differences between tuples and lists. Write a Python program that takes input from the user to define a tuple and finds the sum and mean of the elements in the tuple. Do not use the built-in sum function. [5]

**Question 7.**

- (a) Explain the concept of recursion in Python functions. Write a Python program with a function that accepts a list of numbers from the user, finds the largest number, and returns it. [5]
- (b) Define the `for` loop and the `while` loop in Python with suitable examples. Write a Python program to accept a number from the user and print its multiplication table. Validate that the input is a positive integer. [5]

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